Lesson 6 Test the LCD1602 display

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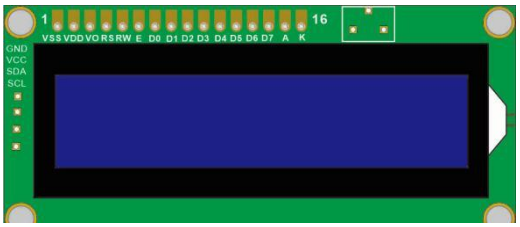
1. Introduction of 1602 LCD

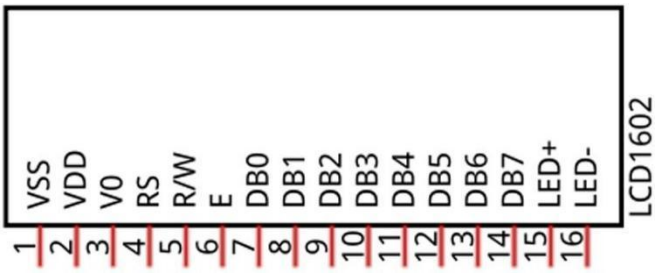
I2C communication

I2C (Inter-Integrated Circuit) is a two-wire serial communication mode, which can be used for the connection of micro controllers and their peripheral equipment. Devices using I2C communication must be connected to the serial data (SDA) line, and serial clock (SCL) line (called I2C bus). Each device has a unique address and can be used as a transmitter or receiver to communicate with devices connected to the bus.

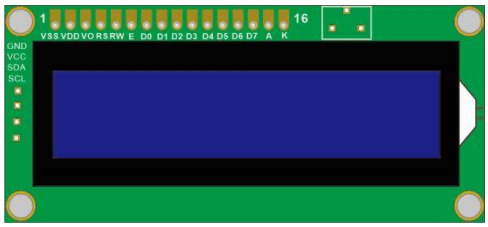
LCD1602 communication

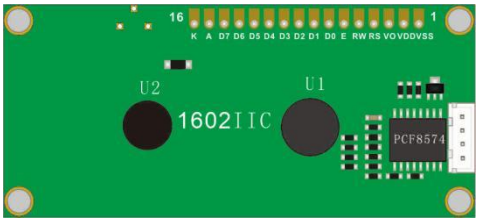
The LCD1602 Display Screen can display 2 lines of characters in 16 columns. It is capable of displaying numbers, letters, symbols, ASCII code and so on. As shown below is a monochrome





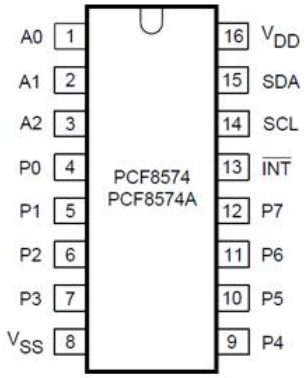
LCD1602 Display Screen along with its circuit pin diagram.
I2C LCD1602 Display Screen integrates an I2C chip, which connects the serial-input & parallel-output circuit to the LCD1602 Display Screen. This allows us to use only 4 lines to the operate the LCD1602.





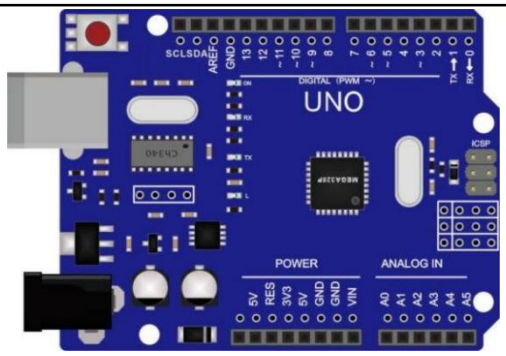
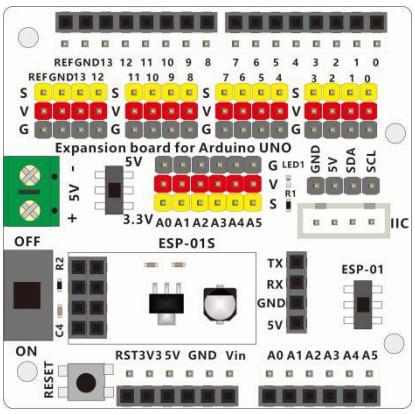
The serial-to-parallel IC chip used in this module is PCF8574, and its default I2C address is 0x27(0x3F).

PCF8574 chip pin diagram:



PCF8574 chip pin and LCD1602 pin are corresponding to each other and connected with each other: So we only need 4 pins to control the 16 pins of the LCD1602 Display Screen through the I2C interface. In this project, we will use the I2C LCD1602 to display some static characters.

2. What do you need to prepare

Components	Quantity	Picture	Remark
A to B USB cable	1		
Arduino UNO R3	1		
Expansion board for Arduino UNO	1		

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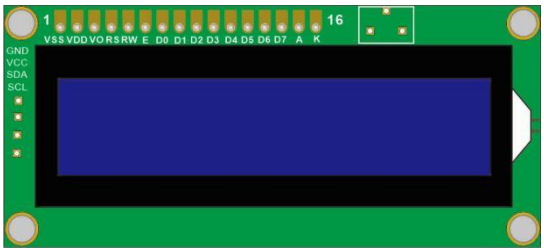
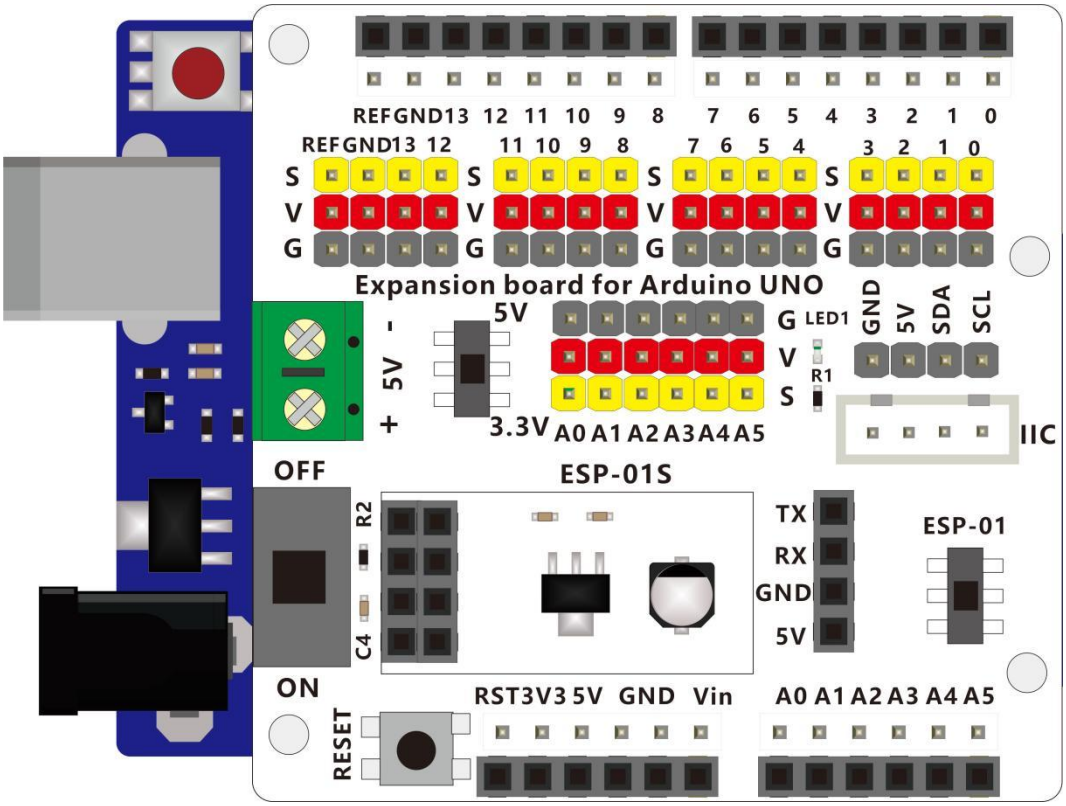
LCD1602 Display Module	1		
4-Pin wires	1		

3. Wiring Circle

Insert the Arduino UNO expansion board onto the Arduino UNO R3 board, paying attention to the assembly direction.As shown in the following figure:

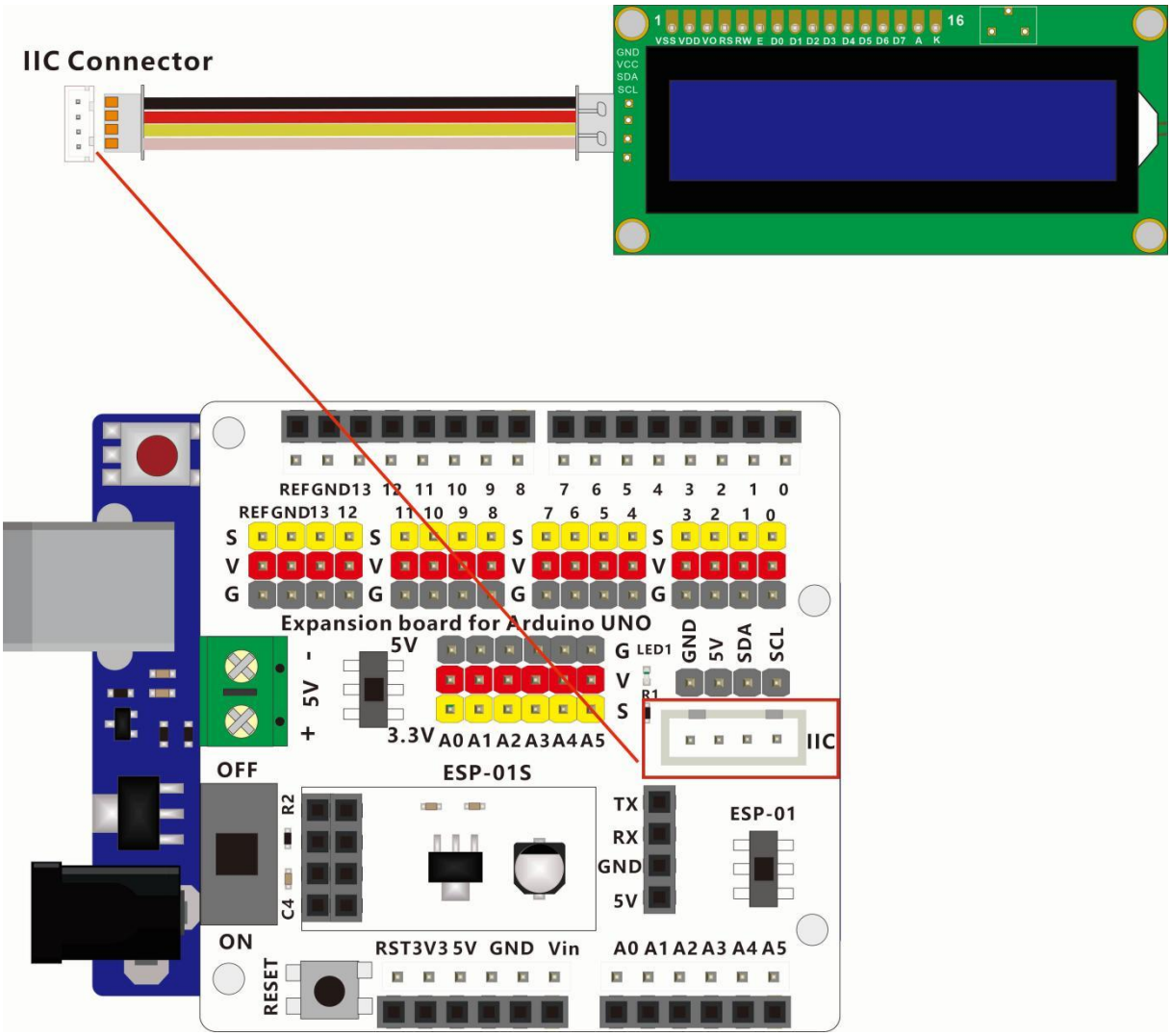


Turn the power switch on the expansion board to ON side and the 5V/3.3V switch to 5V side.
Connect the 1602LCD module to the Arduino UNO Expansion board using the 4 pin wires.
Connect the Arduino UNO to the computer via a A to B USB cable.

Wiring between the LCD1602display and the Arduino UNO expansion board

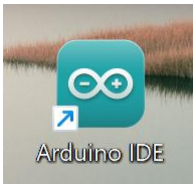
Connector of LCD1602 display	IIC Connector of the orexpansion board
GND	GND
VCC	5V
SDA	SDA
SCL	SCL

The hardware connection diagram is as follows:

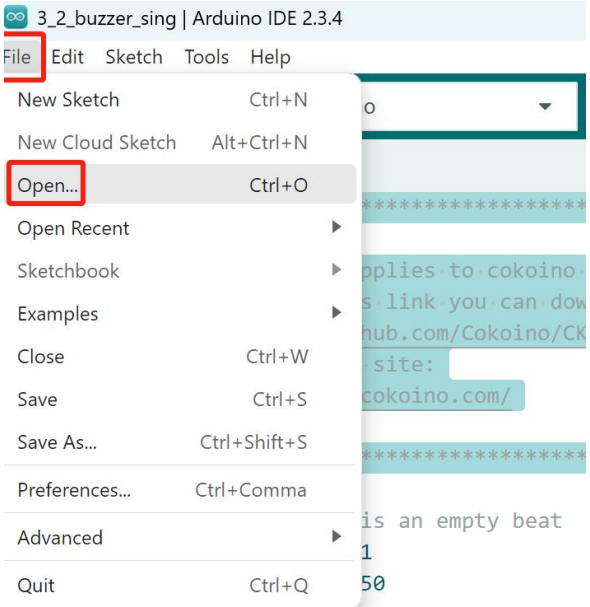


4. Open the test code

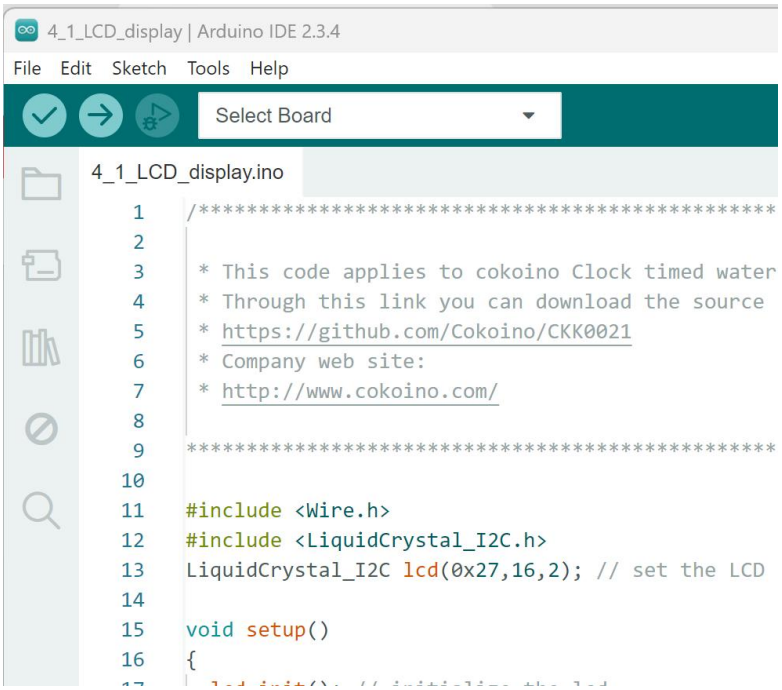
The code used in this lesson is placed in this folder：“ E:\CKK0021-main\Tutorial\sketches”
Double-click the Arduino IDE shortcut on the desktop to open it



Click "File" --- "open"

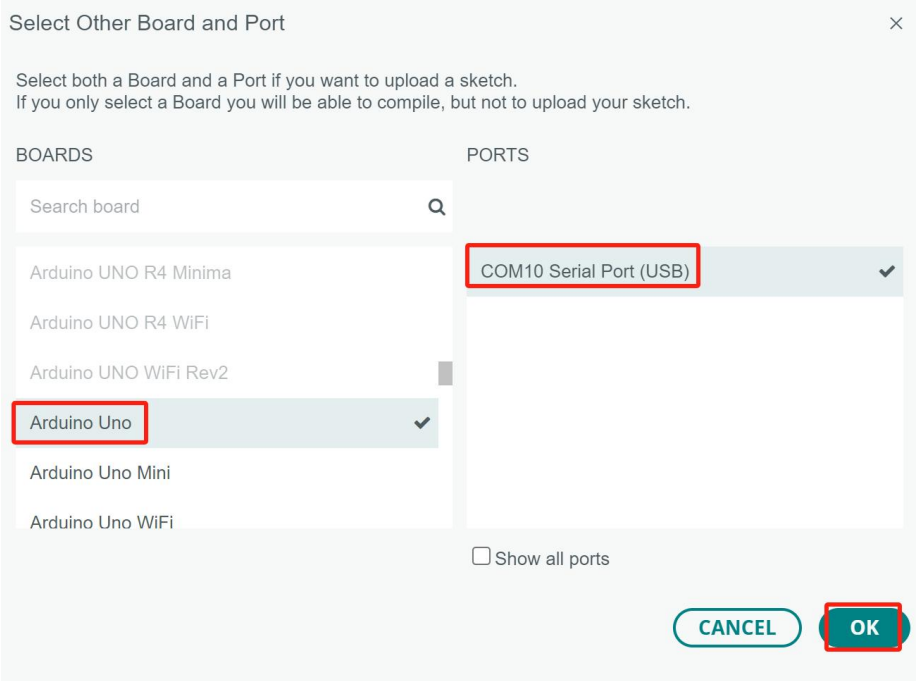


Select the code in the folder named 4_1_LCD_dispaly: E:\COKOINO\CKK0021-main\Tutorial\sketches\4_1_LCD_display,Then click "open".
The code interface after opening is as follows



Select the board "Arduino UNO"and Port "COM10" (COM port is commonly known as an input output port for a device normally PC which enables communication between Arduino

and PC. You can check your arduino com number in device manager, the com port of our arduino board is recognized as COM10 in this tutorial)



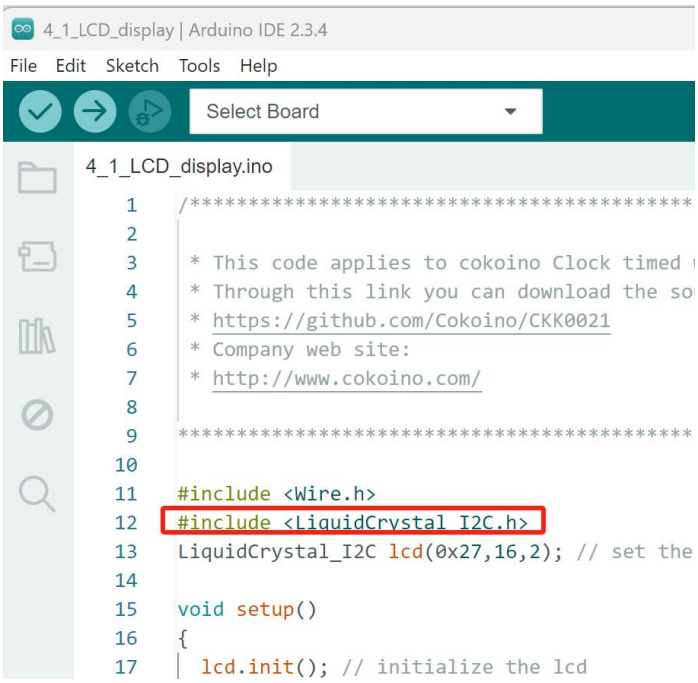
5. Install the LiquidCrystal Library

5.1 Why Use Libraries?

Libraries are incredibly useful when creating a project of any type. They make our development experience much smoother, and there almost an infinite amount out there. They are used to interface with many different sensors, RTCs, Wi-Fi modules, RGB matrices and of course with other components on your board.

5.2 How to use a Library in the sketch

To use a library, you first need to include the library at the top of the sketch. If you find a line of code in the format of **#include <LiquidCrystal_I2C.h>** at the beginning of the code when using our code, it means that you need to add this library file to arduino IDE first before you can successfully upload this code.

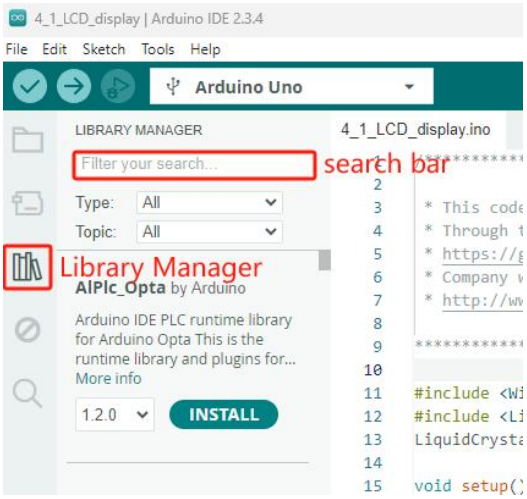


5.3 Install the library to the Arduino IDE

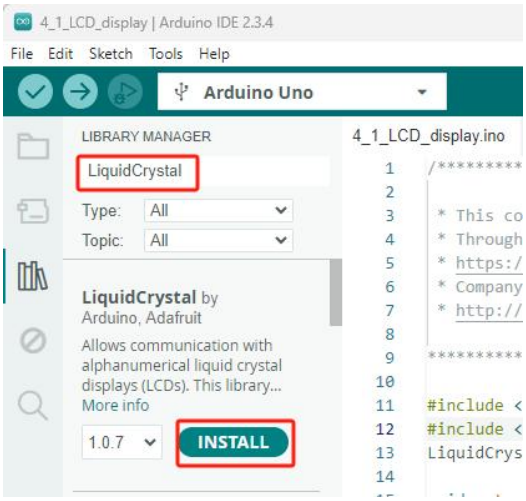
Install the LiquidCrystal library

Method 1：

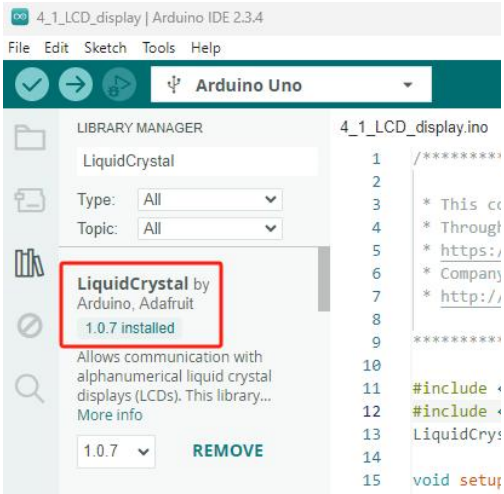
Directly click "Library Manager" on the left side of the Arduino IDE interface, the interface is as follows



Search for “LiquidCrystal” in the search bar, Library Manager will quickly match related library files

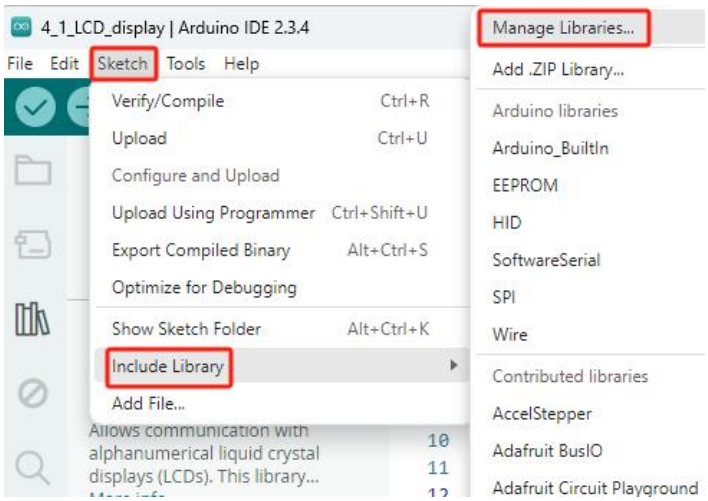


Click "INSTALL" to install the library file of LiquidCrystal. After the installation is complete, it will display "installed"

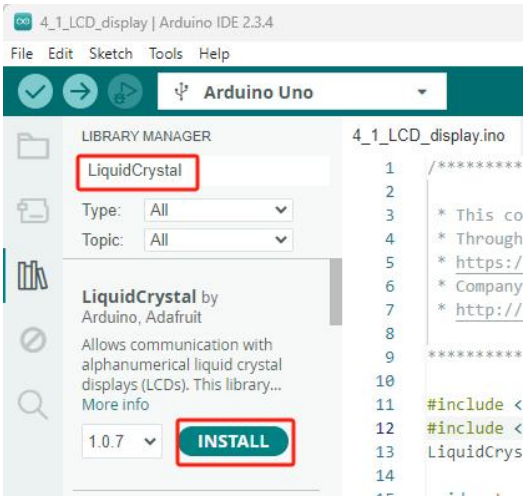


Method 2：

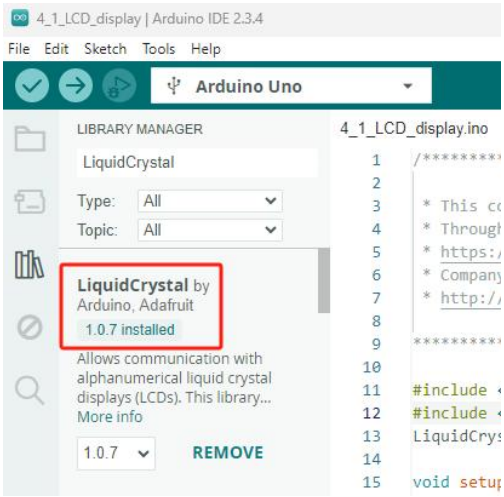
After opening the 4_1_LCD_display code, click "Sketch" --- "Include Library" --- "Manage Libraries..." to open the library management interface.



Search for “LiquidCrystal” in the search bar, Library Manager will quickly match related library files

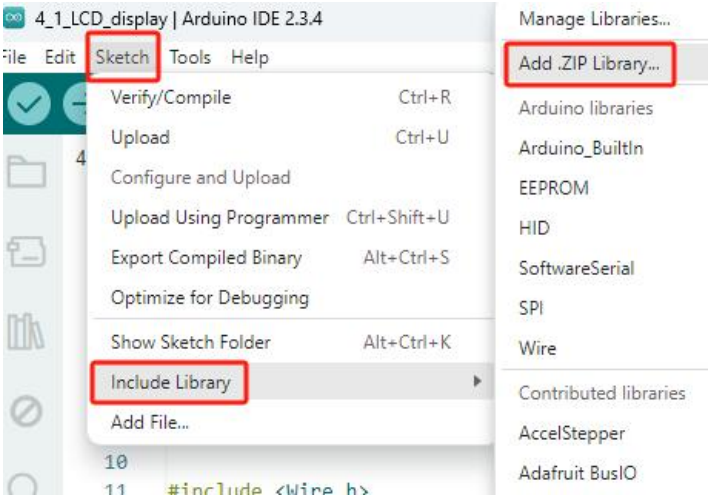


Click "INSTALL" to install the library file of Servo. After the installation is complete, it will display "installed"

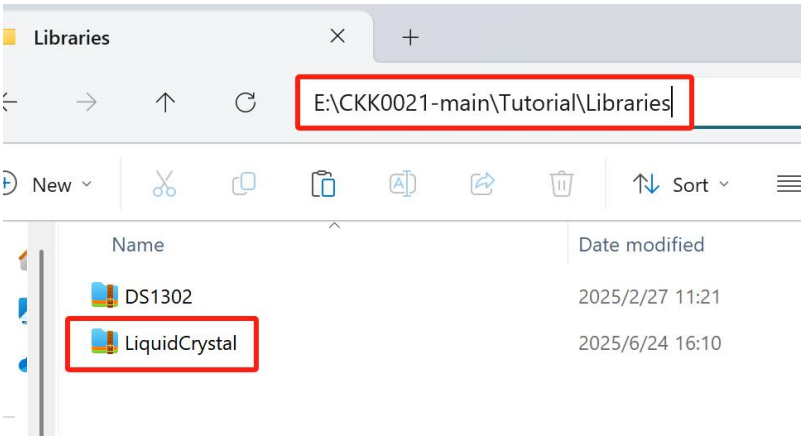


Method 3：

After opening the 4_1_LCD_display code, click "Sketch" ---“Include Library”---“Add.ZIP Library…”



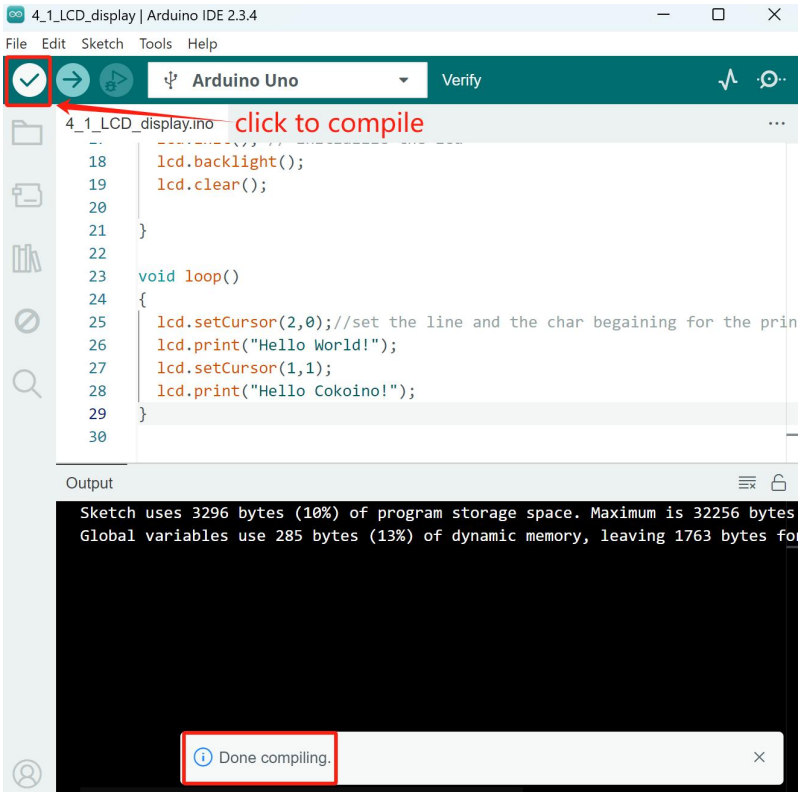
Find the LiquidCrystal.zip file in E:CKK0021-main\Tutorial\Libraries\LiquidCrystal.zip"



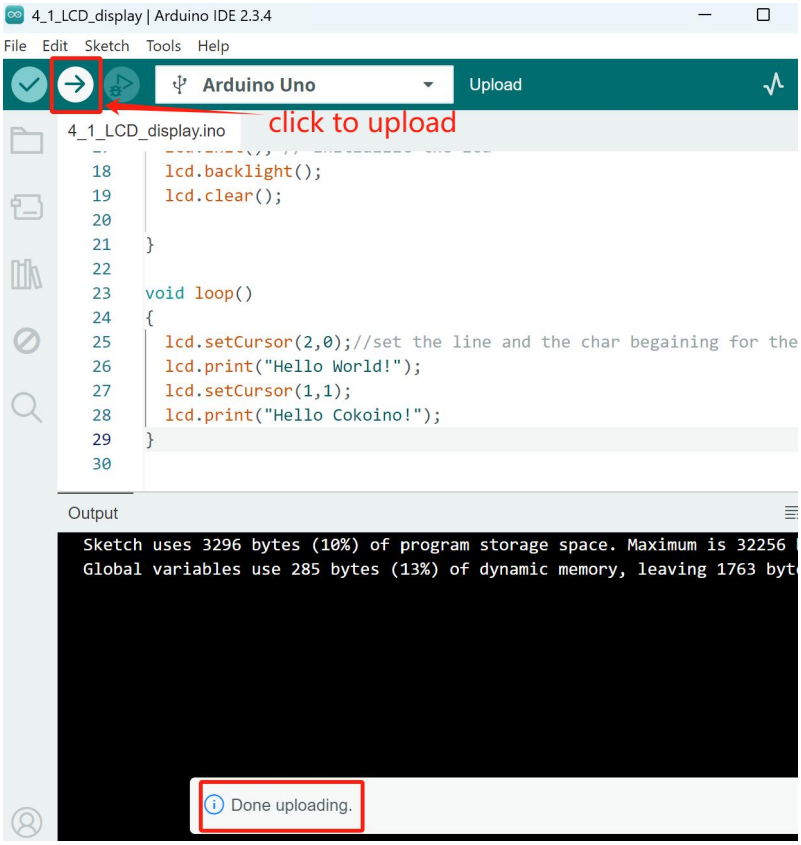
Select the LiquidCrystal.zip file and click "open", and the interface will display "installed" after the library file is installed.

6. Upload the code and test

Click compile button, successfully compiled the code will display "Done compiling"



Click upload button, successfully uploading the code will display "Done uploading". When code is uploaded successfully, the program starts to run.



After the code is uploaded successfully, you can see two lines of text "Hello World!", "Hello Cokoino" displayed on the 1602 LCD display.

7. Code

4_1_LCD_display.ino

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27,16,2); // set the LCD address to 0x27 for a 16 chars and 2 line display

void setup()
{
  lcd.init(); // initialize the lcd
  lcd.backlight();
  lcd.clear();
}

void loop()
{
  lcd.setCursor(2,0);//set the line and the char begaining for the print information
  lcd.print("Hello World!");
  lcd.setCursor(1,1);
  lcd.print("Hello Cokoino!");
}
```

8. Any questions and suggestions are welcome

THANK YOU for participating in this Lesson 4 experience!

If you find any errors, omissions or you have suggestions and/or questions about this lesson, please feel free to contact us:

cokoino@outlook.com

We will make every effort to make changes and correct errors as soon as feasibly possible and publish a revised version.

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